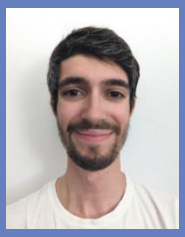
Computer Vision & AI Engineer

Romain GUESDON

+34 623 921 825 | r.guesdon@hotmail.fr | Barcelona (Spain)

archjbald.github.io | linkedin.com/in/romain-guesdon-cv



EXPERIENCE

2024-Present **Technical Lead - FICOSA Automotive** (Spain)

- Developed AI perception systems for European and national collaborative projects:
  - Designed multimodal real-time perception pipelines (cameras, LiDAR, odometry)
  - Implemented, integrated, and evaluated machine learning and deep learning perception models
  - Built annotated multimodal datasets
- Collaborated with cross-functional engineering teams and international partners

2019-2024 **Deep learning and computer vision researcher (PhD) - Univ. Lyon 2 / LIRIS** (France)

Human pose estimation using deep learning: application to passengers in autonomous vehicles

- Designed AI architectures for driver pose estimation
  - Implemented end-to-end training and evaluation pipelines for deep learning models
  - Reduced the synthetic-to-real gap
- Generated synthetic datasets for 3D scene and pose simulation
- Published four papers in international conferences

2020-2023 **University Teacher - Univ. Lyon 2 / LIRIS** (France)

- Taught programming (Python, C++), linear algebra

2019 **R&D Vision Engineer Intern - Siléane** (France, 6 months)

- Developed a C++ vision system for 3D capture of tool manipulation and reproduction with a robot

2017 **Validation Engineer Intern - Schneider Electric** (France, 5 months)

- Developed a Python-based automated firmware validation platform for PLC testing.

EDUCATION

2018 **Exchange semester in Master of Electrical Engineering,** KAIST (Korean Advanced Institute of Science and Technology), South Korea – Information Theory, Computer Vision, Machine Learning

2019 **Master of Engineering (Electrical department)**, INSA Lyon, France – Electronic and Sensors, IT, Machine Learning, Automation, Telecoms

SUMMARY

Computer Vision Engineer, PhD, specialized in camera calibration, multi-view/multi-sensor perception and robustness to domain shift, with 2D/3D deep learning systems in industrial projects.

SKILLS

Computer Vision

- Camera Calibration
- Multi-view Geometry
- Homography
- Sensor Fusion

ML

- Detection & Tracking
- Pose Estimation
- Fine-tuning
- Data Augmentation

Programming

- Python, C++, MATLAB

Frameworks & Tools

- PyTorch, OpenCV, CUDA
- ONNX, FastAPI, MLflow
- Docker, Git, Linux

LANGUAGES

- English: fluent (C1)
- Spanish: upper intermediate (B2)
- French: native
- German: intermediate (B1)